

Section : Common

Q.1 If $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ then the matrix AB is equal to

1. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$
2. $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$
3. $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$
4. $\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

Options

1. 1
2. 2
3. 3
4. 4

Q.2 $\int_1^4 |x - 2| dx$ is equal to

1. 5
2. $\frac{7}{2}$
3. $\frac{3}{2}$
4. $\frac{5}{2}$

Options

1. 1
2. 2
3. 3
4. 4

Q.3 The solution of the differential equation $\log_e \left(\frac{dy}{dx} \right) = 3x + 4y$ is given by

1. $4e^{3x} + 3e^{-4y} + C = 0$, where C is constant of integration
2. $3e^{3x} + 4e^{-4y} + C = 0$, where C is constant of integration
3. $4e^{-3x} + 3e^{4y} + C = 0$, where C is constant of integration
4. $3e^{-3x} + 4e^{4y} + C = 0$, where C is constant of integration

Options

1. 1
2. 2
3. 3
4. 4

Q.4 Let $A = [a_{ij}]_{n \times n}$ be a matrix. Then

Match List-I with List-II

List-I List-II

(A) $A^T = A$ (I) A is a singular matrix

(B) $A^T = -A$ (II) A is a non-singular matrix

(C) $|A| = 0$ (III) A is a skew symmetric matrix

(D) $|A| \neq 0$ (IV) A is a symmetric matrix

Choose the correct answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

2. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

3. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

4. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)

Options 1. 1

2. 2

3. 3

4. 4

Q.5

If $y = 3e^{2x} + 2e^{3x}$, then $\frac{d^2y}{dx^2} + 6y$ is equal to

1. $\frac{dy}{dx}$

2. $5\frac{dy}{dx}$

3. $6\frac{dy}{dx}$

4. $30\frac{dy}{dx}$

Options 1. 1

2. 2

3. 3

4. 4

Q.6

The probability distribution of a random variable X is given by

X	0	1	2
P(X)	$1 - 7a^2$	$\frac{1}{2}a$	$\frac{1}{4}a^2$

If $a > 0$, then $P(0 < X \leq 2)$ is equal to

1. $\frac{1}{16}$

2. $\frac{3}{18}$

3. $\frac{7}{16}$

4. $\frac{9}{16}$

Options 1. 1

2. 2

3. 3

4. 4

Q.7

The area (in sq. units) of the region bounded by the parabola $y^2 = 4x$ and the line $x = 1$ is

1. $\frac{1}{3}$

2. $\frac{4}{3}$

3. $\frac{5}{3}$

4. $\frac{8}{3}$

Options 1. 1

2. 2

3. 3

4. 4

Q.8 The interval, on which the function $f(x) = x^2 e^{-x}$ is increasing, is equal to

1. $(-\infty, \infty)$

2. $(-\infty, 2) \cup (2, \infty)$

3. $(-2, 0)$

4. $(0, 2)$

Options 1. 1

2. 2

3. 3

4. 4

Q.9 Consider the LPP: Minimize $Z = x + 2y$ subject to $2x + y \geq 3, x + 2y \geq 6, x, y \geq 0$. The optimal feasible solution occurs at

1. $(6, 0)$ only

2. $(0, 3)$ only

3. Neither $(6, 0)$ nor $(0, 3)$

4. Both $(6, 0)$ and $(0, 3)$

Options 1. 1

2. 2

3. 3

4. 4

Q.10 If A is a square matrix and I is the identity matrix of same order such that $A^2 = I$, then $(A - I)^3 + (A + I)^3 - 3A$ is equal to

1. A

2. $2A$

3. $3A$

4. $5A$

Options 1. 1

2. 2

3. 3

4. 4

Q.11 If $A = \begin{bmatrix} 0 & 0 & \sqrt{3} \\ 0 & \sqrt{3} & 0 \\ \sqrt{3} & 0 & 0 \end{bmatrix}$, then $|\text{adj } A|$ is equal to

1. 3

2. 9

3. 27

4. 81

Options 1. 1

2. 2

3. 3

4. 4

Q.12 If the maximum value of the function $f(x) = \frac{\log_e x}{x}, x > 0$ occurs at $x = a$, then $a^2 f''(a)$ is equal to

1. $-\frac{5}{e}$

2. $-\frac{1}{e}$

3. $-\frac{1}{e^3}$

4. $-5e^3$

Options 1. 1

2. 2

3. 3

4. 4

Q.13 The corner points of the feasible region associated with the LPP: Maximise $Z = px + qy, p, q > 0$ subject to $2x + y \leq 10, x + 3y \leq 15, x, y \geq 0$ are $(0, 0), (5, 0), (3, 4)$ and $(0, 5)$. If optimum value occurs at both $(3, 4)$ and $(0, 5)$, then

1. $p = q$

2. $p = 2q$

3. $p = 3q$

4. $q = 3p$

Options 1. 1

2. 2

3. 3

4. 4

Q.14 The integral $I = \int \frac{e^{5 \log_e x} - e^{4 \log_e x}}{e^{3 \log_e x} - e^{2 \log_e x}} dx$ is equal to

1. $\frac{x}{2} + C$, where C is the constant of integration

2. $\frac{x^2}{2} + C$, where C is the constant of integration

3. $\frac{x^3}{3} + C$, where C is the constant of integration

4. $\frac{x^4}{4} + C$, where C is the constant of integration

Options 1. 1

2. 2

3. 3

4. 4

Q.15 Which of the following are linear first order differential equations?

(A) $\frac{dy}{dx} + P(x)y = Q(x)$

(B) $\frac{dx}{dy} + P(y)x = Q(y)$

(C) $(x - y)\frac{dy}{dx} = x + 2y$

(D) $(1 + x^2)\frac{dy}{dx} + 2xy = 2$

Choose the \textbf{correct} answer from the options given below:

1. (A), (B) and (D) only

2. (A) and (B) only

3. (A), (B) and (C) only

4. (A), (B), (C) and (D)

Options 1. 1

2. 2

3. 3

4. 4

Q.16 The probability distribution of the random variable X is given by

X 0 1 2 3

P(X) 0.2 k 2k 2k

The variance of the random variable X is given by

1. $\frac{764}{625}$

2. $\frac{1}{625}$

3. 1

4. $\frac{108}{25}$

Options 1. 1

2. 2

3. 3

4. 4

Q.17 Which of the following inequalities holds true?

(A) $\sqrt{5} + \sqrt{3} > \sqrt{6} + \sqrt{2}$.

(B) If $a > b$ and $c < 0$, then $\frac{a}{c} < \frac{b}{c}$.

(C) $\frac{1}{x^2} > \frac{1}{x} > 1$, if $0 < x < 1$.

(D) If a and b are positive integers and $\frac{a-b}{6.25} = \frac{4}{2.5}$, then $b > a$.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only

2. (A), (B) and (C) only

3. (A) and (B) only

4. (B) and (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.18 Let A be a non-singular matrix of order 3 and $|A| = 15$, then $|\text{adj } A|$ is equal to

1. 15

2. 45

3. 225

4. 150

Options 1. 1

2. 2

3. 3

4. 4

Q.19 If a 95% confidence interval for a population mean was reported to be 132 to 160 and sample standard deviation $\sigma = 50$, then the size of the sample in the study is:
(Given $Z_{0.025} = 1.96$)

1. 90

2. 95

3. 50

4. 49

Options 1. 1

2. 2

3. 3

4. 4

Q.20 If P , Q and R are three singular matrices given by

$$P = \begin{bmatrix} 2 & 3a \\ 4 & 3 \end{bmatrix}, Q = \begin{bmatrix} b & 5 \\ 2a & 6 \end{bmatrix}$$

and $R = \begin{bmatrix} a^2 + b^2 - c & 1 - c \\ c + 1 & c \end{bmatrix}$, then the value of $(2a + 6b + 17c)$ is

1. 30
2. 18
3. 34
4. 24

Options 1. 1

2. 2
3. 3
4. 4

Q.21 Let $F(z)$ be the cumulative density function of the standard normal variate Z , then which of the following are correct?

(A) $F(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^z e^{-t^2/2} dt, -\infty < Z < \infty$

(B) $F(-Z) = 1 - F(Z)$

(C) $F(0) = 0$

(D) $F(\infty) = 1$

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (C) and (D) only
4. (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.22 The original value of an asset minus the accumulated depreciation at a given date is known as

1. Salvage value
2. Book value
3. Scrap value
4. Lost Value

Options 1. 1

2. 2
3. 3
4. 4

Q.23 If $y = a + b(x - 2022)$ is a straight line trend using the least square method for the following data

Year(x)	2020	2021	2022	2023	2024
Profit(Rs. '000)(y)	2	3	4	5	2

Then the value of $\frac{a}{b}$ is:

1. 15
2. 5
3. 16
4. $\frac{2}{3}$

Options 1. 1

2. 2
3. 3
4. 4

Q.24 The slope of the normal to the curve $y = 2x^2$ at $x = 1$ is:

1. 4
2. -4
3. $\frac{1}{2}$
4. $-\frac{1}{4}$

Options 1. 1

2. 2

3. 3

4. 4

Q.25 If $X = 11$ and $Y = 3$, then $X \bmod Y = (X + aY) \bmod Y$ holds

1. Only for even integral values of a

2. Only for odd integral values of a

3. For all integral values of a

4. For $a = 0$ only

Options 1. 1

2. 2

3. 3

4. 4

Q.26

Let $e^{\alpha y} + e^{\beta y} + \gamma x^2 + \delta \log |x| + C = 0$, where $C \in \mathbb{R}$, $\lfloor \cdot \rfloor$ be a particular solution of the differential equation $x(e^{2y} - 1) dy + (x^2 - 1)e^y dx = 0$ and passes through the point $(1, 1)$.

The value of $(\alpha + \beta + \gamma + \delta - C)$ is

1. $e - 1$

2. $e^2 - 1$

3. $e + \frac{1}{e}$

4. $\frac{1}{e}$

Options 1. 1

2. 2

3. 3

4. 4

Q.27 The total cost $C(x)$ in Rupees associated with the production of x units of an item is given by

$$C(x) = 0.007x^3 + 26x^2 + 15x + 400.$$

The marginal cost when 10 items are produced is:

1. ₹ 537.1

2. ₹ 441.15

3. ₹ 1575

4. ₹ 875.25

Options 1. 1

2. 2

3. 3

4. 4

Q.28 How many minimum number of times must a man toss a fair coin so that the probability of having at least one head is more than 90%?

1. 3

2. 4

3. 5

4. 10

Options 1. 1

2. 2

3. 3

4. 4

Q.29 If $A = \begin{bmatrix} 3 & 7 \\ 4 & -2 \end{bmatrix}$, $X = \begin{bmatrix} \alpha \\ -2 \end{bmatrix}$, $B = \begin{bmatrix} 7 \\ 32 \end{bmatrix}$
and $AX = B$, then the value of α is

1. 7
2. $\frac{4}{3}$
3. 1
4. 5

Options 1. 1

2. 2
3. 3
4. 4

Q.30 A person invested ₹ 10000 in a stock of a company for 6 years.
The value of his investment at the end of each year is given below:

2018	2019	2020	2021	2022	2023
₹11000	₹11500	₹13000	₹11800	₹12200	₹14000

The compound annual growth rate (CAGR) of his investment is:

[Given $(1.4)^{\frac{1}{6}} = 1.058$]

1. 5.8%
2. 4.2%
3. 6.8%
4. 3.2%

Options 1. 1

2. 2
3. 3
4. 4

Q.31 If $e^y = \log x$, then which of the following is true?

1. $x \frac{d^2y}{dx^2} - \left(\frac{dy}{dx} \right)^2 + \frac{dy}{dx} = 0$
2. $\frac{d^2y}{dx^2} - x \frac{dy}{dx} = 0$
3. $\frac{d^2y}{dx^2} - \left(\frac{dy}{dx} \right)^2 + 1 = 0$
4. $x \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 + \frac{dy}{dx} = 0$

Options 1. 1

2. 2
3. 3
4. 4

Q.32 A sofa set costing ₹ 36000 has a useful life of 10 years. If the annual depreciation is ₹ 3000, then the scrap value by linear method is:

1. ₹ 4000
2. ₹ 6000
3. ₹ 4200
4. ₹ 5400

Options 1. 1

2. 2
3. 3
4. 4

Q.33 Find the least non-negative remainder obtained when 3^{128} is divided by 7.

1. 2
2. 3
3. 4
4. 5

Options 1. 1

2. 2
3. 3
4. 4

Q.34 Match list-I with List -II

List-I

(A) An observed set of population selected for analysis

(B) A specific characteristic of the population

(C) A specific characteristic of the sample

(D) A statement made about a population parameter for testing

List-II

(I) Parameter

(II) Hypothesis

(III) Statistic

(IV) Sample

Choose the correct answer from the options given below:

1. (A) – (I), (B) – (II), (C) – (III), (D) – (IV)

2. (A) – (IV), (B) – (III), (C) – (I), (D) – (II)

3. (A) – (I), (B) – (II), (C) – (IV), (D) – (III)

4. (A) – (IV), (B) – (I), (C) – (III), (D) – (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.35 For the given 5 values, 15, 18, 21, 27, 39; the three year moving averages are:

1. 18, 21, 29

2. 18, 22, 29

3. 18, 23, 37

4. 18, 20, 28

Options 1. 1

2. 2

3. 3

4. 4

Q.36 An annuity in which the periodic payment begin on a fixed date and continue forever is called

1. Sinking Fund

2. Perpetuity

3. Coupon payment

4. Bond

Options 1. 1

2. 2

3. 3

4. 4

Q.37 Two runners, Ajay and Vijay complete a 600 m race in 38 seconds and 48 seconds respectively. By how many meters will Ajay defeat Vijay?

1. 120 m

2. 140 m

3. 125 m

4. 50 m

Options 1. 1

2. 2

3. 3

4. 4

Q.38 If $\int \frac{1+x \log x}{x e^{-x}} dx = e^x f(x) + C$,

where C is constant of integration, then $f(x)$ is

1. e^x

2. $\log x$

3. $\frac{1}{x}$

4. $\frac{1}{x^2}$

Options 1. 1

2. 2

3. 3

4. 4

Q.39 Which of the following is NOT a basic requirement of the linear programming problem (LPP)?

1. All the elements of an LPP should be quantifiable.

2. All decision variables should assume non-negative values.

3. There are a finite number of decision variables and a finite number of constraints.

4. It deals with optimizing number of objectives more than one.

Options 1. 1

2. 2

3. 3

4. 4

Q.40 Which of the following are the assumptions underlying the use of t-distribution?

(A) The variance of population is known.

(B) The samples are drawn from a normally distributed population.

(C) Sample standard deviation is an unbiased estimate of the population variance.

(D) It depends on a parameter known as degree of freedom.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only

2. (A), (B) and (C) only

3. (B) and (D) only

4. (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.41 Which of the following are correct about the Sinking Fund?

(A) It is a fixed term account.

(B) It is a set-up for a particular upcoming expense.

(C) A fixed amount at regular intervals is deposited in the Sinking Fund.

(D) It can be used in any emergency.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only

2. (A), (B) and (C) only

3. (C) and (D) only

4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.42 What is the mean of the numbers obtained on throwing a die having written 1 on three faces, 2 on two faces and 5 on 1 face?

1. 2

2. 4

3. 1

4. 8

Options 1. 1

2. 2

3. 3

4. 4

Q.43 Which of the following is not a component of the time series?

1. Trend component

2. Cyclical Component

3. Seasonal Component

4. Average Component

Options 1. 1

2. 2

3. 3

4. 4

Match **List-I** with **List-II**

List-I	List-II
(Matrix)	(Inverse of the Matrix)
(A) $\begin{bmatrix} 1 & 7 \\ 4 & -2 \end{bmatrix}$	(I) $\begin{bmatrix} 2/15 & 1/10 \\ -1/15 & 1/5 \end{bmatrix}$
(B) $\begin{bmatrix} 6 & -3 \\ 2 & 4 \end{bmatrix}$	(II) $\begin{bmatrix} 1/5 & -2/15 \\ -1/10 & 7/30 \end{bmatrix}$
(C) $\begin{bmatrix} 5 & 2 \\ -5 & 4 \end{bmatrix}$	(III) $\begin{bmatrix} 1/15 & 7/30 \\ 2/15 & -1/30 \end{bmatrix}$
(D) $\begin{bmatrix} 7 & 4 \\ 3 & 6 \end{bmatrix}$	(IV) $\begin{bmatrix} 2/15 & -1/15 \\ 1/6 & 1/6 \end{bmatrix}$

Choose the **correct** answer from the options given below:

- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
- (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
- (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.45 A tub contains 60 litres of milk. From this tub, 6 litres of milk was taken out and replaced with water. This whole process was repeated further two more times. How much milk is there in the tub now?

- 29.16 litre
- 43.74 litre
- 42.24 litre
- 38.74 litre

Options 1. 1

2. 2

3. 3

4. 4

Q.46 A person wishes to purchase a house for ₹ 39,65,000 with a down payment of ₹ 5,00,000 and balance in equal monthly installments (EMI) for 25 years.

If bank charges 6% per annum compounded monthly, then EMI on reducing balance payment method is:

[Given $(1.005)^{300} = 4.465$]

- ₹ 22325
- ₹ 36542
- ₹ 21652
- ₹ 34500

Options 1. 1

2. 2

3. 3

4. 4

Q.47 Which of the following are correct?

- Time series analysis does not help to understand the behavior of a variable in the past.
- Time series predict the future behavior of variable.
- Time series helps to plan future operations.

(D) The main aim of the time series analysis is to derive conclusions after arranging the time series in a systematic manner.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (B), (C) and (D) only
4. (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.48 Which of the following statements is incorrect?

1. If two rows or two columns of a determinant are identical, then the value of the determinant is zero.
2. If all the elements in any one row of the determinant are zero, then the determinant value is zero.
3. The value of the determinant remains unchanged if its rows and columns are interchanged.
4. If any two rows of a determinant are interchanged, then the sign of the determinant remains unchanged.

Options 1. 1

2. 2

3. 3

4. 4

Q.49 Which of the following statements are correct in reference to the linear programming problem (LPP):

Maximize $Z = 5x + 2y$

subject to the following constraints

$3x + 5y \leq 15, 5x + 2y \leq 10, x \geq 0, y \geq 0$

- (A) The LPP has a unique optimal solution at (2, 0) only.
(B) The feasible region is bounded with corner points (0, 0), (2, 0), (20/19, 45/19) and (0, 3).
(C) The optimal value is unique, but there are an infinite number of optimal solutions.
(D) The feasible region is unbounded.

Choose the correct answer from the options given below:

1. (A) and (D) only
2. (A), (B) and (C) only
3. (A), (C) and (D) only
4. (B) and (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.50 A person can row a boat in still water at the rate of 5 km/hr. It takes him 4 times as long to row upstream of a river as to row downstream to cover same distance in the same river. The speed of flow of the stream is

1. 5 km/hr
2. 3 km/hr
3. 6.5 km/hr
4. 4 km/hr

Options 1. 1

2. 2

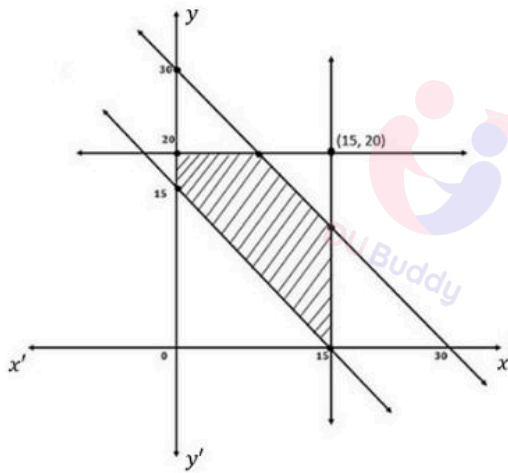
3. 3

4. 4

Section : Core Mathematics

Q.51

Which one of the following set of constraints does the given shaded region represent?



1. $x + y \leq 30, x + y \geq 15, x \leq 15, y \leq 20, x, y \geq 0$
2. $x + y \leq 30, x + y \geq 15, y \leq 15, x \leq 20, x, y \geq 0$
3. $x + y \geq 30, x + y \leq 15, x \leq 15, y \leq 20, x, y \geq 0$
4. $x + y \geq 30, x + y \leq 15, y \leq 15, x \leq 20, x, y \geq 0$

Options 1. 1

2. 2

3. 3

4. 4

Q.52

Consider the line $\vec{r} = \hat{i} - 2\hat{j} + 4\hat{k} + \lambda(-\hat{i} + 2\hat{j} - 4\hat{k})$

Match \textbf{List-I} with \textbf{List-II}

\textbf{List-I}

\textbf{List-II}

(A) A point on the given line

(I) $(\frac{-1}{\sqrt{21}}, \frac{2}{\sqrt{21}}, \frac{-4}{\sqrt{21}})$

(B) direction ratios of the line

(II) $(4, -2, -2)$

(C) direction cosines of the line

(III) $(1, -2, 4)$

(D) direction ratios of a line perpendicular to given line (IV) $(-1, 2, -4)$

Choose the \textbf{correct} answer from the options given below:

1. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
2. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
3. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
4. (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.53

Let $A = [a_{ij}]_{2 \times 3}$ and $B = [b_{ij}]_{3 \times 2}$, then $|5AB|$ is equal to

1. $5^2 \cdot |A| \cdot |B|$

2. $5^3 \cdot |A| \cdot |B|$

3. $5^2 |AB|$

4. $5^3 |AB|$

Options 1. 1

2. 2
3. 3
4. 4

Q.54 The area (in sq. units) of the region bounded by the curve $y = x^5$, the x-axis and the ordinates $x = -1$ and $x = 1$ is equal to

1. $\frac{1}{6}$
2. $\frac{1}{3}$
3. $\frac{1}{2}$
4. $\frac{2}{3}$

Options 1. 1
2. 2
3. 3
4. 4

Q.55 Let $A = \{1, 2, 3\}$. Then, the number of relations containing $(1, 2)$ and $(1, 3)$, which are reflexive and symmetric but not transitive, is

1. 1
2. 2
3. 3
4. 4

Options 1. 1
2. 2
3. 3
4. 4

Q.56 Consider the differential equation, $x \frac{dy}{dx} = y(\log_e y - \log_e x + 1)$, then which of the following are true?

- (A) It is a linear differential equation
- (B) It is a homogenous differential equation
- (C) Its general solution is $\log_e\left(\frac{y}{x}\right) = Cx$, where C is constant of integration
- (D) Its general solution is $\log_e\left(\frac{x}{y}\right) = Cy$, where C is constant of integration
- (E) If $y(1) = 1$, then its particular solution is $y = x$

Choose the \textbf{correct} answer from the options given below:

1. (A), (D) and (E) only
2. (A) and (D) only
3. (B) and (C) only
4. (B), (C) and (E) only

Options 1. 1
2. 2
3. 3
4. 4

Q.57 If the function $f(x) = \begin{cases} \frac{k \cos x}{\pi - 2x} & : x \neq \frac{\pi}{2} \\ 3 & : x = \frac{\pi}{2} \end{cases}$ is continuous at $x = \frac{\pi}{2}$, then k is equal to

1. 6
2. 5
3. -6
4. 4

Options 1. 1
2. 2
3. 3

Q.58 The corner points of the feasible region of the LPP: Minimize $Z = -50x + 20y$ subject to $2x - y \geq -5$, $3x + y \geq 3$, $2x - 3y \leq 12$ and $x, y \geq 0$ are

1. $(0, 5), (0, 6), (1, 0), (6, 0)$
2. $(0, 3), (0, 5), (3, 0), (6, 0)$
3. $(0, 3), (0, 5), (1, 0), (6, 0)$
4. $(0, 5), (0, 6), (1, 0), (3, 0)$

Options 1. 1

2. 2
3. 3
4. 4

Q.59 for $|x| < 1$, $\sin(\tan^{-1} x)$ equal to

1. $\frac{1}{\sqrt{1+x^2}}$
2. $\frac{1}{\sqrt{1-x^2}}$
3. $\frac{x}{\sqrt{1-x^2}}$
4. $\frac{x}{\sqrt{1+x^2}}$

Options 1. 1

2. 2
3. 3
4. 4

Q.60 The function $f(x) = \tan x - x$

1. is a decreasing function on $[0, \frac{\pi}{2})$
2. is an increasing function on $[0, \frac{\pi}{2})$
3. is a constant function
4. is neither increasing nor decreasing function on $[0, \frac{\pi}{2})$

Options 1. 1

2. 2
3. 3
4. 4

Q.61 If A and B are skew-symmetric matrices, then which one of the following is NOT true?

1. $A^3 + B^5$ is skew-symmetric
2. A^{19} is skew-symmetric
3. B^{14} is symmetric
4. $A^4 + B^5$ is symmetric

Options 1. 1

2. 2
3. 3
4. 4

Q.62 Match \textbf{List-I} with \textbf{List-II}

\textbf{List-I} \hspace{10em} \textbf{List-II}

\textbf{Definite integral} \hspace{10em} \textbf{Value}

(A) $\int_0^1 \frac{2x}{1+x^2} dx$ (I) 2

(B) $\int_{-1}^1 \sin^3 x \cos^4 x dx$ (II) $\log_e(\frac{3}{2})$

(C) $\int_0^\pi \sin x dx$ (III) $\log_e 2$

(D) $\int_2^3 \frac{2}{x^2-1} dx$ (IV) 0

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
3. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options

1. 1
2. 2
3. 3
4. 4

Q.63 Let $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. If $A^T + A = I$, then

1. $\theta = 2n\pi + \frac{\pi}{3}, n \in Z$
2. $\theta = n\pi, n \in Z$
3. $\theta = (2n+1)\frac{\pi}{2}, n \in Z$
4. $\theta = 2n\pi + \frac{\pi}{6}, n \in Z$

Options

1. 1
2. 2
3. 3
4. 4

Q.64 The rate of change of area of a circle with respect to its circumference when radius is 4cm, is

1. $2 \text{ cm}^2/\text{cm}$
2. $4 \text{ cm}^2/\text{cm}$
3. $8 \text{ cm}^2/\text{cm}$
4. $16 \text{ cm}^2/\text{cm}$

Options

1. 1
2. 2
3. 3
4. 4

Q.65 Let $AX = B$ be a system of three linear equations in three variables. Then the system has

- (A) a unique solutions if $|A| = 0$
- (B) a unique solutions if $|A| \neq 0$
- (C) no solutions if $|A| = 0$ and $(\text{adj } A) B \neq 0$
- (D) infinitely many solutions if $|A| = 0$ and $(\text{adj } A) B = 0$

Choose the correct answer from the options given below:

1. (A), (C) and (D) only
2. (B), (C) and (D) only
3. (B) only

4. (B) and (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.66 If A is any event associated with sample space and If E_1, E_2, E_3 are mutually exclusive and exhaustive events. Then which of the following are true?

(A) $P(A) = P(E_1)P(E_1|A) + P(E_2)P(E_2|A) + P(E_3)P(E_3|A)$

(B) $P(A) = P(A|E_1)P(E_1) + P(A|E_2)P(E_2) + P(A|E_3)P(E_3)$

(C) $P(E_i|A) = \frac{P(A|E_i)P(E_i)}{\sum_{i=1}^3 P(A|E_i)P(E_i)}, i = 1, 2, 3$

(D) $P(A|E_i) = \frac{P(E_i|A)P(E_i)}{\sum_{i=1}^3 P(E_i|A)P(E_i)}, i = 1, 2, 3$

Choose the correct answer from the options given below:

1. (A) and (C) only

2. (A) and (D) only

3. (B) and (D) only

4. (B) and (C) only

Options 1. 1

2. 2

3. 3

4. 4

Q.67 If $\hat{i}, \hat{j}$ and $\hat{k}$ are unit vectors along co-ordinates axes OX, OY and OZ respectively, then which of the following is/are true?

(A) $\hat{i} \times \hat{i} = \vec{0}$

(B) $\hat{i} \times \hat{k} = \hat{j}$

(C) $\hat{i} \cdot \hat{i} = 1$

(D) $\hat{i} \cdot \hat{j} = 0$

Choose the correct answer from the options given below:

1. (A) and (B) only

2. (A), (C) and (D) only

3. (A) only

4. (A), (B), (C) and (D)

Options 1. 1

2. 2

3. 3

4. 4

Q.68 The integral $I = \int e^x \left(\frac{x-1}{3x^2} \right) dx$ is equal to

1. $\frac{1}{3} \left(\frac{x^2}{2} - x \right) + C$, where C is constant of integration

2. $\left(\frac{x^2}{2} - x \right) e^x + C$, where C is constant of integration

3. $\frac{1}{3x^2} e^x + C$, where C is constant of integration

4. $\frac{1}{3x} e^x + C$, where C is constant of integration

Options 1. 1

2. 2

3. 3

4. 4

Q.69 If a line makes angles α, β, γ with the positive directions of x-axis, y-axis and z-axis respectively, then $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma$ is equal to

1. 1
2. 2
3. 3
4. -2

Options

1. 1
2. 2
3. 3
4. 4

Q.70 Let $f : R \rightarrow R$ be defined as $f(x) = 10x$. Then (Where R is the set of real numbers)

1. f is both one-one and onto
2. f is onto but not one-one
3. f is one-one but not onto
4. f is neither one-one nor onto

Options

1. 1
2. 2
3. 3
4. 4

Q.71 Let $\vec{a} = \hat{i} + 4\hat{j}$, $\vec{b} = 4\hat{j} + \hat{k}$ and $\vec{c} = \hat{i} - 2\hat{k}$. If $\vec{d}$ is a vector perpendicular to both $\vec{a}$ and $\vec{b}$ such that $\vec{c} \cdot \vec{d} = 16$, then $|\vec{d}|$ is equal to

1. $\sqrt{33}$
2. $2\sqrt{33}$
3. $3\sqrt{33}$
4. $4\sqrt{33}$

Options

1. 1
2. 2
3. 3
4. 4

Q.72 The integrating factor of the differential equation $(x \log_e x) \frac{dy}{dx} + y = 2 \log_e x$ is

1. $\log_e x$
2. x
3. $\frac{1}{x}$
4. $\frac{1}{\log_e x}$

Options

1. 1
2. 2
3. 3
4. 4

Q.73 Match List-I with List-II

Let A and B are two events such that $P(A) = 0.8$, $P(B) = 0.5$, $P(B|A) = 0.4$

List-I **List-II**

- (A) $P(A \cap B)$ (I) 0.2
- (B) $P(A|B)$ (II) 0.32
- (C) $P(A \cup B)$ (III) 0.64
- (D) $P(A')$ (IV) 0.98

Choose the correct answer from the options given below:

1. (A) - (II), (B) - (IV), (C) - (III), (D) - (I)
2. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
3. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)
4. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

Options 1. 1

2. 2
3. 3
4. 4

Q.74 If A and B are invertible matrices then which of the following statement is NOT correct?

1. $\text{adj}A = |A|A^{-1}$
2. $(A + B)^{-1} = A^{-1} + B^{-1}$
3. $|A^{-1}| = |A|^{-1}$
4. $(AB)^{-1} = B^{-1}A^{-1}$

Options 1. 1

2. 2
3. 3
4. 4

Q.75

Let $A = \begin{bmatrix} 1 & 2 & 1 \\ -1 & 3 & 2 \\ 2 & 4 & 1 \end{bmatrix}$ and M_{ij} , A_{ij} respectively denote the minor, co-factor of an element a_{ij} of matrix A, then which of the following are true?

- (A) $M_{22} = -1$
- (B) $A_{23} = 0$
- (C) $A_{32} = 3$
- (D) $M_{23} = 1$
- (E) $M_{32} = -3$

Choose the correct answer from the options given below:

1. (A) and (B) only
2. (A), (B), (C) and (E) only
3. (A), (D) and (E) only
4. (A), (C) and (E) only

Options 1. 1

2. 2
3. 3
4. 4

Q.76 Match List-I with List-II

List-I

List-II

- (A) $f(x) = |x|$ (I) Not differentiable at $x = -2$ only
- (B) $f(x) = |x + 2|$ (II) Not differentiable at $x = 0$ only
- (C) $f(x) = |x^2 - 4|$ (III) Not differentiable at $x = 2$ only
- (D) $f(x) = |x - 2|$ (IV) Not differentiable at $x = 2, -2$ only

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
3. (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
4. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

Options 1. 1

2. 2
3. 3
4. 4

Q.77 A black and a red die are rolled simultaneously. The probability of obtaining a sum greater than 9, given that the black

resulted in a 5 is

1. $1/2$
2. 1
3. $2/3$
4. $1/3$

Options

1. 1
2. 2
3. 3
4. 4

Q.78

The area (in sq. units) of the region bounded by $y = 2\sqrt{1-x^2}$, $x \in [0, 1]$ and x-axis is equal to

1. 1
2. 2
3. $\frac{\pi}{2}$
4. $\frac{\pi}{4}$

Options

1. 1
2. 2
3. 3
4. 4

Q.79

If the points A, B, C with position vectors $20\hat{i} + \lambda\hat{j}$, $5\hat{i} - \hat{j}$ and $10\hat{i} - 13\hat{j}$ respectively are collinear, then the value of λ is

1. 12
2. -37
3. 37
4. -12

Options

1. 1
2. 2
3. 3
4. 4

Q.80

Let $y = \sin(\cos x^2)$, then the value of $\frac{dy}{dx}$ at $x = \frac{\sqrt{\pi}}{2}$ is equal to

1. $-\frac{\sqrt{\pi}}{2} \cos\left(\frac{1}{\sqrt{2}}\right)$
2. $-\sqrt{\frac{\pi}{2}} \cos\left(\frac{1}{\sqrt{2}}\right)$
3. $-\sqrt{\frac{\pi}{2}} \sin\left(\frac{1}{\sqrt{2}}\right)$
4. $\sqrt{\frac{\pi}{2}} \sin\left(\frac{1}{\sqrt{2}}\right)$

Options

1. 1
2. 2
3. 3
4. 4

Q.81 Match List-I with List-II

List-I

List-II

- | | |
|--|---------|
| (A) The minimum value of $f(x) = (2x - 1)^2 + 3$ | (I) 4 |
| (B) The maximum value of $f(x) = - x + 1 + 4$ | (II) 10 |
| (C) The minimum value of $f(x) = \sin(2x) + 6$ | (III) 3 |

(D) The maximum value of $f(x) = -(x-1)^2 + 10$ (IV) 5

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)
3. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)
4. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.82 $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\tan x}{\tan x + \cot x} dx$ is equal to

1. $\frac{\pi}{4}$
2. 0
3. $\frac{\pi}{6}$
4. $\frac{\pi}{12}$

Options 1. 1

2. 2
3. 3
4. 4

Q.83 If $\vec{a} + \vec{b} + \vec{c} = \vec{0}$ and $|\vec{a}| = 3$, $|\vec{b}| = 5$, $|\vec{c}| = 7$, then the angle between $\vec{a}$ and $\vec{b}$ is

1. $\frac{\pi}{2}$
2. $\frac{\pi}{3}$
3. $\frac{\pi}{4}$
4. $\frac{\pi}{6}$

Options 1. 1

2. 2
3. 3
4. 4

Q.84 If A and B are any two events such that $P(B) = P(A \text{ and } B)$, then which of the following is correct

1. $P(B|A) = 1$
2. $P(A|B) = 1$
3. $P(B|A) = 0$
4. $P(A|B) = 0$

Options 1. 1

2. 2
3. 3
4. 4

Q.85 The shortest distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{4} = \frac{y-4}{6} = \frac{z-5}{8}$ is equal to

1. 0

2. $\sqrt{\frac{29}{5}}$

3. $\sqrt{\frac{5}{29}}$

4. $\sqrt{5}$

Options

1. 1

2. 2

3. 3

4. 4